Exploring Sleep Health and Lifestyle Factors for Sleep Disorder Classification Using Machine Learning

AI 221 - University of the Philippines Diliman

1 Deomary Angelo Franco 2 Jovie Ruth Mendoza 3 Ely Jun Pates

***Abstract*—This study investigates the classification and clustering of sleep disorders using supervised and unsupervised learning techniques, leveraging a dataset on sleep health and lifestyle.**

I. Background of the Study

Sleep is an essential physiological process that maintains both physical and mental well-being. Quality sleep supports various bodily functions, including memory consolidation, immune system strengthening, and emotional regulation. On the other hand, its absence can lead to a range of health issues, including cognitive impairments, mood disorders, cardiovascular problems, and weakened immune responses. With this mentioned, understanding the factors that influence sleep quality is crucial for improving overall health outcomes.

Lifestyle factors, such as diet, physical activity, and stress levels, have been shown to have a significant impact on sleep quality. Low levels of these factors can be a cause of disrupted sleep patterns and contribute to the development of sleep disorders.

Among the various sleep disorders, insomnia and sleep apnea are two of the most prevalent and debilitating conditions [2]. Insomnia is the presence of a long sleep latency, frequent nocturnal awakenings, or prolonged periods of wakefulness during the sleep period or even frequent transient arousals [1]. Sleep apnea, on the other hand, the health condition where airflow either ceases or decreases episodically during sleep [3]. This involves the obstruction of the airway during sleep, leading to intermittent breathing interruptions, oxygen deprivation, and frequent awakenings.

Despite the widespread impact of these sleep disorders, accurately diagnosing them and identifying the most effective treatment approaches remains a complex task. The increasing availability of large-scale health data provides an opportunity to explore machine learning methods for predicting and classifying sleep disorders, including insomnia, sleep apnea, and their comorbidities. By leveraging data-driven approaches, we can gain valuable insights into the underlying patterns of sleep health and improve the precision of sleep disorder diagnoses.

This study aims to investigate the use of machine learning techniques to predict sleep disorders, specifically insomnia and sleep apnea. The focus will be on exploring supervised learning models for classification, as well as unsupervised learning methods for clustering, to better understand the relationships within the dataset and enhance the diagnostic accuracy for sleep-related conditions.

II. Statement of the Problem

The goal of this study is to predict sleep disorders (Insomnia, Sleep Apnea, or None) based on various lifestyle and health features using supervised learning classification algorithms. Furthermore, the study aims to explore unsupervised learning for clustering similar patterns in the dataset without the use of labels.

III. Objectives of the Study

This study aims to develop a model that can predict sleep disorders, including Insomnia, and Sleep Apnea, based on various health and lifestyle features. This objective is addressed through two main approaches:

A. Supervised Learning

By applying multiple classification algorithms (such as Logistic Regression and hyperparameters), the study aims to compare performances and identify the model that achieves the highest accuracy to be fine-tuned for performance improvement.

B. Unsupervised Learning

The study also aims to explore clustering techniques using algorithms like KModes, to uncover hidden patterns within the data without using the label of ”Sleep Disorder”.

IV. Methodology

A. Data Preprocessing

Missing values in the dataset were handled by filling them with ”No Sleep Disorder” for the target variable. This ensures that no data points are lost during the analysis.

The Blood Pressure feature was split into two separate columns: bp-systolic and bp-diastolic, as blood pressure is typically recorded in these two distinct values.

Irrelevant features, such as Person ID, were dropped to avoid unnecessary noise during model training. The dataset was randomly split into training and test subsets at a ratio of 60:40. The splitting process ensures that 60 percent of the data was used for training the machine learning models, while the remaining 40 percent was held out for testing the performance of the models.

The test data was stratified according to the target variable, "Sleep Disorder", ensuring that the proportions of the different classes (i.e., None, Insomnia, and Sleep Apnea) were preserved in the test set to maintain the distribution of the classes in both the training and testing sets, making the evaluation more representative of the actual data distribution.

B. Exploratory Data Analysis

This was performed to check the distribution of features across different classes of the target variable. Visualizations were used to explore the relationships between features (blood pressure, age, lifestyle factors, etc.) and the target (Insomnia, Sleep Apnea, No Disorder). This will used in determining which features are more important for classification and which are less informative.

C. Classification using Supervised Learning

Supervised learning involves training a model using labeled data to predict the target variable. In this case, the target variable is the sleep disorder which has values Insomnia, Sleep Apnea, or None.

For both the baseline logistic regression and optimized logistic regression models, a simple pipeline was created to streamline the preprocessing and modeling workflow. The pipeline consists of two main steps:

One Hot Encoding: Categorical variables in the dataset (such as "gender", "marital status", etc.) were transformed into numerical format using One Hot Encoding. This technique creates binary columns for each category in the original feature, enabling the algorithm to work with categorical data.

Standard Scaling: Continuous numerical variables (such as "age", "sleep duration", etc.) were scaled using the StandardScaler. This scaler standardizes the data by removing the mean and scaling to unit variance, ensuring that all features contribute equally to the model's performance without being influenced by differing magnitudes or units.

The following algorithms were utilized:

1) Baseline Logistic Regression: As a baseline, Logistic Regression will first define a model that will not require tuning of hyperparameters. Preprocessing of data is simply applying a OneHotEncoder for the categorical data and StandardScaler for the numerical data. "Fig. 2"

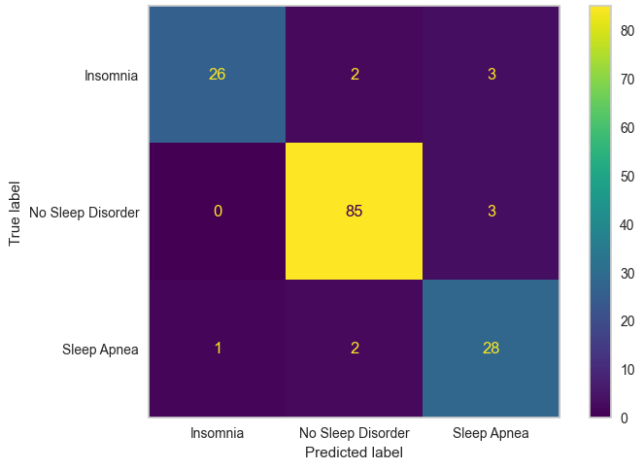

Fig. 1. Correlation Matrix in Logistic Regression Analysis

2) Optimized Logistic Regression: Hyperparameter Tuning using RandomSearchCV is used to maximize model performance, improve accuracy, precision, and recall for the sleep disorder classification. The model has retained its accuracy at 92.67 with misclassifying 11 out of 150. This preprocessing is the pipeline stated in the Data Preprocessing Pipeline.

3) Pycaret Classification: This algorithm used to automatically select the most relevant features for classification to reduce overfitting and see if the model's accuracy improves. We have selected the top 3 models (other than a logistic regression model) and tune them. We used Light Gradient Boosting Machine as model 1, Ridge Classifier as model 2, and Extreme Gradient Boosting as model 3 (See Fig. 3 for Confusion Matrices of these models). We will also stack the outputs of these models and generate stacked model. The summary of hyperparameters, accuracy, and weighted f1 were given on Tables I and II.

TABLE I
SUMMARY OF HYPERPARAMETERS USED ON MODELS

	Hyperparameters
Logistic Regression Base	{'C': 1.0, 'class_weight': None, 'dual': False...
Logistic Regression Tuned	{'C': 0.31489116479568624, 'class_weight': Non
Light Gradient Boosting Machine	{'boosting_type': 'gbdt', 'class_weight': None...
Ridge Classifier	{'alpha': 6.78, 'class_weight': None, 'copy_X'...
Extreme Gradient Boosting	{'objective': 'multi:softprob', 'base_score': ...

TABLE II
ACCURACY AND WEIGHTED F1

	Accuracy	Weighted f1
Logistic Regression Base	0.926667	0.926804
Logistic Regression Tuned	0.926667	0.926833
Light Gradient Boosting Machine	0.9	0.898127
Ridge Classifier	0.893333	0.894519
Extreme Gradient Boosting	0.933333	0.932891
Pycaret Model Stacked Tuned	0.926667	0.926181

4) XGBoost: Being known for its high performance and efficiency, XGB algorithm using default hyperpa-

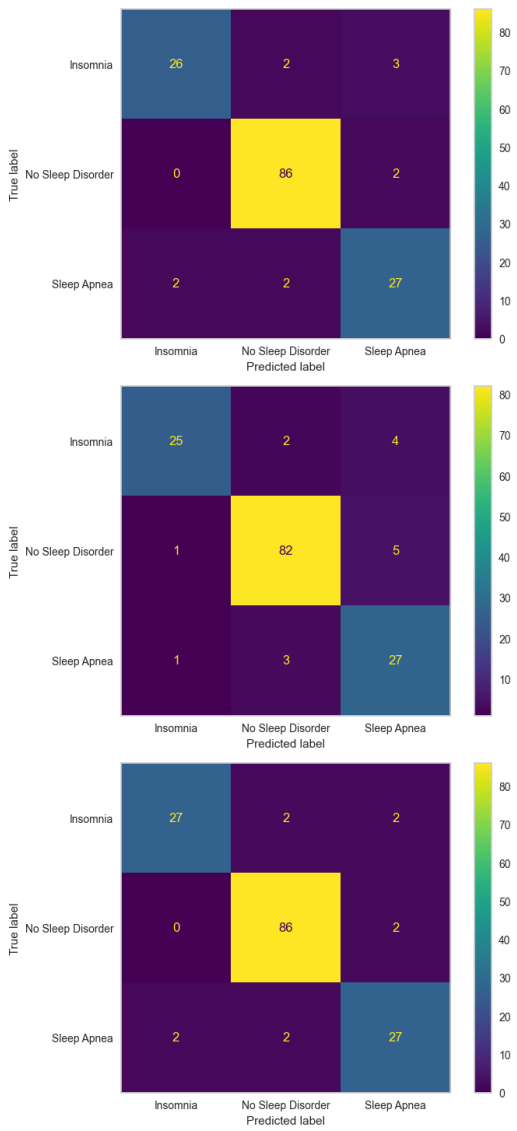

Fig. 2. Correlation Matrix: Model 1, 2, and 3

rameters was explored to identify the best-performing model.

D. Clustering using Unsupervised Learning

Unsupervised clustering methods were applied to the dataset, where the target variable was ignored. The methods used for this analysis is the KModes Clustering.

1) K-Modes Clustering: For this study, this technique was used to group the data points (i.e., individuals in your study) into distinct clusters based on their categorical attributes, such as lifestyle, health behaviors, and demographic information uncovering hidden patterns or similarities in the dataset that were not obvious, like common traits among people with certain sleep disorders.

An Elbow Plot is used to visualize the relationship between the number of clusters (K) and the distortion score. (See Fig. 5)

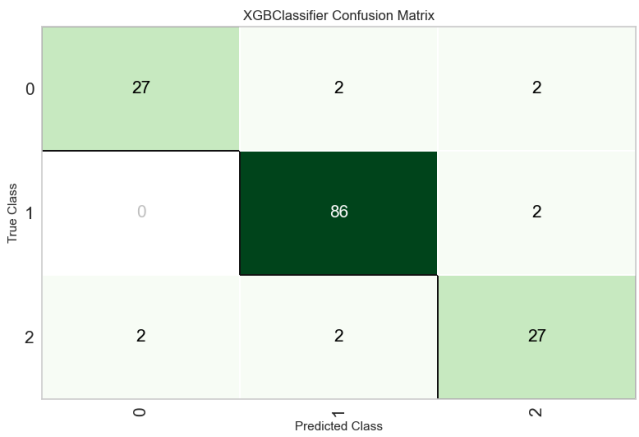

Fig. 3. Confusion Matrix: XGB Classifier

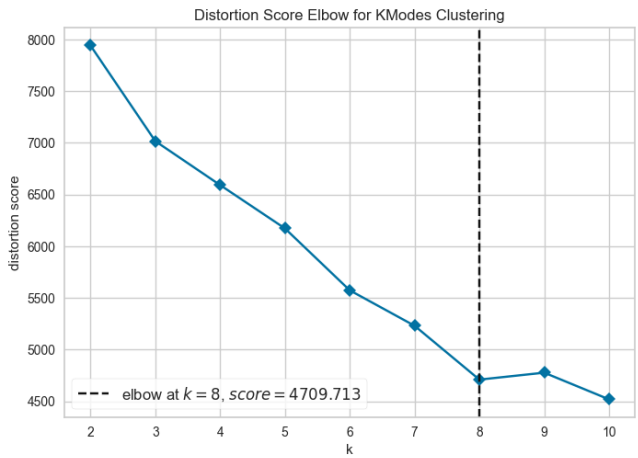

Fig. 4. Elbow Plot: K vs. Distortion Score

To evaluate how well each data point fits within its assigned cluster compared to other clusters, we have used Silhouette Plot. This will help determine whether the clusters are distinct and well-separated (higher silhouette scores (close to +1), indicates well-separated clusters). (See Fig. 5)

V. Results

The initial model was evaluated on the test dataset and resulted to an accuracy of 92.67 percent. Out of 150 test samples, 11 were misclassified, demonstrating a strong performance in identifying sleep disorders. This baseline result set the stage for comparing the effectiveness of other classification algorithms, including XGBoost, which achieved the highest accuracy among all models tested.

XGBoost, which got the highest accuracy all other models, hacing an accuracy of 93.33 percent, had only 10 misclassifications out of 150 test samples. Its high performance highlights its ability to handle complex interactions effectively between features and capture non-linear relationships, making it the most suitable algorithm for this classification task.

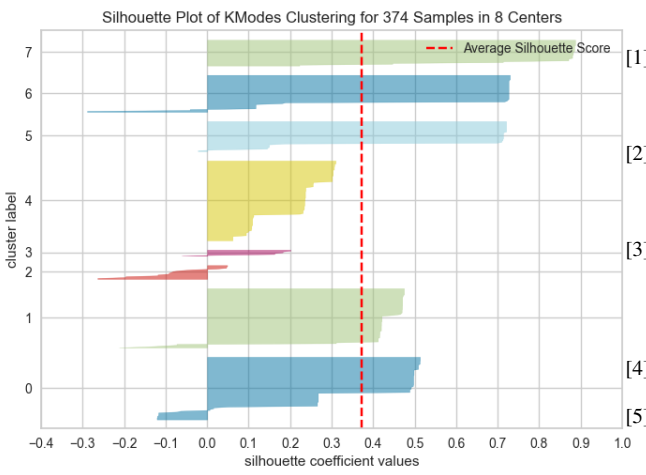

Fig. 5. Silhouette Plot of 374 Samples

Additionally, K-Modes Clustering was used as an unsupervised learning approach to uncover patterns and groupings in the dataset without utilizing the ”Sleep Disorder” label. The optimal number of clusters was determined using the Elbow Plot, which suggested 8 clusters. The analysis revealed distinct groups based on demographic, lifestyle, and health-related features: (See Appendix C)

The combination of supervised and unsupervised learning approaches allowed for a comprehensive analysis of the dataset. XGBoost proved most effective in classifying sleep disorders with labeled data, K-Modes Clustering created another perspective by identifying hidden patterns and similarities among individuals, offering potential directions for further research and intervention strategies.

VI. Summary

The study employed both supervised and unsupervised learning approaches to analyze sleep disorders using health and lifestyle data. For supervised learning, Logistic Regression served as the baseline model, achieving an accuracy of 92.67 percent with 11 misclassifications out of 150 test samples. However, XGBoost emerged as the best-performing model, achieving the highest accuracy of 93.33 with only 10 misclassifications. This result highlights XGBoost’s ability to capture complex, non-linear relationships in the dataset, making it the most effective algorithm for classifying sleep disorders.

In addition to classification, K-Modes Clustering was utilized as an unsupervised learning method to uncover hidden patterns in the dataset without using the ”Sleep Disorder” label. The optimal number of clusters, determined through the Elbow Plot, was 8.

By combining the predictive power of XGBoost with the exploratory insights provided by K-Modes Clustering, the study offered a comprehensive understanding of sleep disorders.

References

[1] Roth T. (2007). Insomnia: definition, prevalence, etiology, and consequences. Journal of clinical sleep medicine : JCSM : official publication of the American Academy of Sleep Medicine, 3(5 Suppl), S7–S10.

[2] Ong, J. C., Crawford, M. R., and Wallace, D. M. (2021). Sleep Apnea and Insomnia: Emerging Evidence for Effective Clinical Management. Chest, 159(5), 2020–2028. https://doi.org/10.1016/j.chest.2020.12.002

[3] Abd-Alrazaq, A., Aslam, H., AlSaad, R., Alsahli, M., Ahmed, A., Damseh, R., Aziz, S., and Sheikh, J. (2024). Detection of Sleep Apnea Using Wearable AI: Systematic Review and Meta-Analysis. Journal of medical Internet research, 26, e58187. https://doi.org/10.2196/58187

[4] PyCaret. (2020, April). https://pycaret.org/. PyCaret Version 1.0.0

[5] Ali, M. (2020, April). PyCaret: An open-source, low-code machine learning library in Python (Version 1.0.0). https://www.pycaret.org.

[6] Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., and Duchesnay, E. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825–2830.

APPENDIX

APPENDIX A

Histogram plot of All Data, No Sleep Disorder, Sleep Apnea, Insomnia

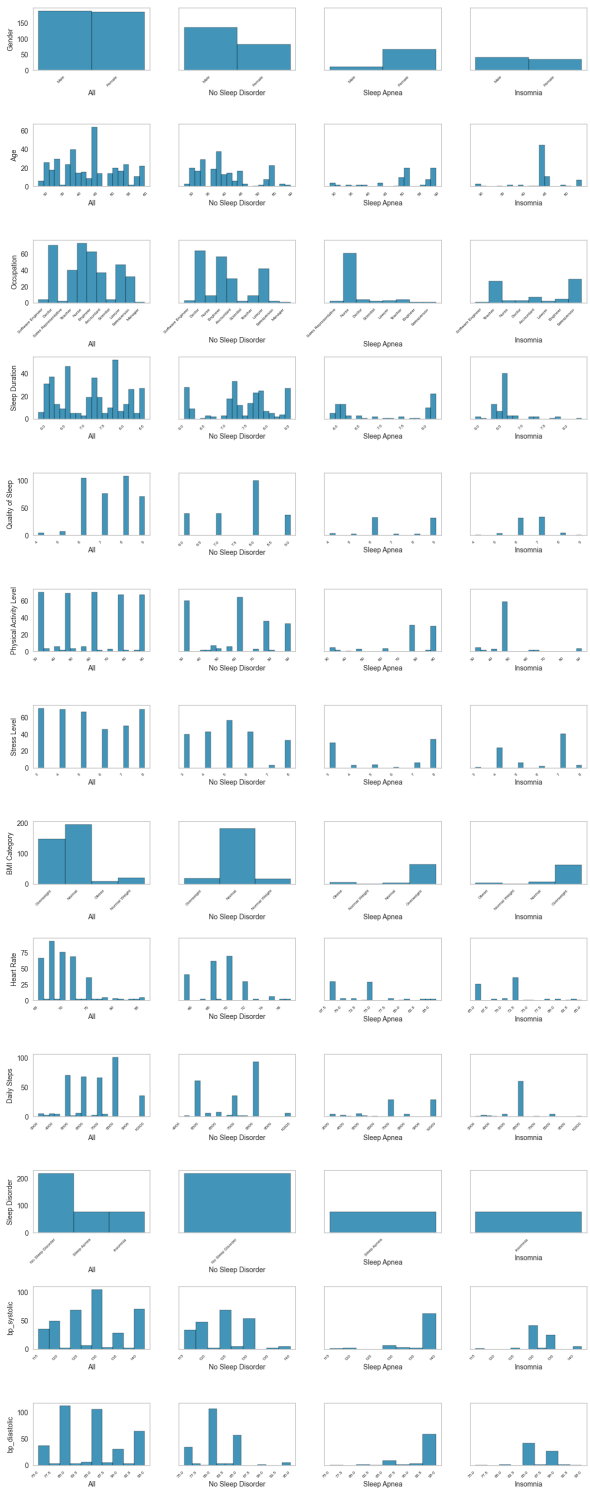

Fig. 6. Histogram plot of All Data, No Sleep Disorder, Sleep Apnea, Insomnia

APPENDIX B

Aggregated Mean of Features per Cluster

	Cluster 0	Cluster 1	Cluster 2	Cluster 3
Gender_Female	-0.96	0.98	0.78	-0.99
Gender_Male	0.96	-0.98	-0.78	0.99
Occupation_Accountant	-0.33	-0.33	0.46	-0.33
Occupation_Doctor	-0.17	-0.48	-0.18	0.79
Occupation_Engineer	0.24	-0.45	-0.14	-0.45
Occupation_Lawyer	1.41	-0.38	-0.20	-0.38
Occupation_Nurse	-0.49	1.92	0.25	-0.49
Occupation_Salesperson	-0.31	-0.31	-0.31	-0.31
Occupation_Teacher	-0.35	-0.35	-0.35	0.06
Occupation_infrequent_sklearn	-0.01	0.08	0.87	2.05
BMI Category_Normal	0.82	-1.04	-1.04	-1.04
BMI Category_Normal Weight	-0.13	-0.18	3.33	-0.24
BMI Category_Obese	0.00	-0.17	-0.17	6.03
BMI Category_Overweight	-0.78	1.21	-0.45	-0.81
Sleep Disorder_Insomnia	-0.38	-0.47	-0.22	0.73
Sleep Disorder_No Sleep Disorder	0.62	-0.96	0.36	-1.19
Sleep Disorder_Sleep Apnea	-0.38	1.63	-0.22	0.72
Age	-0.40	1.18	-0.30	-0.44
Sleep Duration	0.54	-0.11	-0.08	-0.36
Quality of Sleep	0.45	0.06	-0.26	-0.89
Physical Activity Level	0.76	0.99	-0.08	-0.26
Stress Level	-0.19	0.10	0.08	0.28
Heart Rate	-0.13	0.43	0.69	3.44
Daily Steps	0.64	0.84	-0.33	-2.14
bp_systolic	-0.04	1.35	-0.65	1.51
bp_diastolic	-0.08	1.54	-0.56	0.99

	Cluster 4	Cluster 5	Cluster 6	Cluster 7
Gender_Female	-0.31	0.84	-0.63	0.75
Gender_Male	0.31	-0.84	0.63	-0.75
Occupation_Accountant	-0.33	2.18	0.13	-0.33
Occupation_Doctor	1.04	-0.48	-0.48	-0.48
Occupation_Engineer	0.63	-0.30	-0.33	-0.45
Occupation_Lawyer	-0.38	-0.38	-0.24	-0.38
Occupation_Nurse	-0.49	-0.42	-0.49	-0.49
Occupation_Salesperson	-0.31	-0.31	2.29	-0.31
Occupation_Teacher	-0.35	0.19	-0.20	2.89
Occupation_infrequent_sklearn	-0.17	-0.17	-0.17	-0.17
BMI Category_Normal	0.96	0.74	-1.04	-1.04
BMI Category_Normal Weight	-0.24	0.24	-0.24	-0.24
BMI Category_Obese	-0.17	-0.17	-0.17	-0.17
BMI Category_Overweight	-0.81	-0.81	1.24	1.24
Sleep Disorder_Insomnia	-0.46	-0.44	1.57	1.57
Sleep Disorder_No Sleep Disorder	0.75	0.73	-1.00	-0.93
Sleep Disorder_Sleep Apnea	-0.46	-0.44	-0.35	-0.43
Age	-0.39	-0.63	0.26	0.12
Sleep Duration	0.27	0.10	-0.81	-0.67
Quality of Sleep	0.07	0.57	-0.95	-0.37
Physical Activity Level	-0.75	0.05	-0.66	-0.77
Stress Level	0.09	-0.75	0.89	-0.56
Heart Rate	-0.37	-0.51	0.59	-1.03
Daily Steps	-0.49	0.13	-0.58	-0.57
bp_systolic	-0.66	-1.69	0.16	0.72
bp_diastolic	-0.76	-1.50	0.04	0.74

Appendix C
Clusters

No Sleep Disorder	Physically Active Males	
	Normal Weight Females	
	Physically Inactive but Healthy Individuals	
	Young Female Accountants	
Sleep Apnea	Old Overweight Female Nurses	
Insomnia	Male Salesperson	
	Female Teachers	
N/A	Male with High Blood Pressure	

Fig. 7. Clusters Generated

Appendix D
Baseline Logistic Regression Hyperparameters

	Baseline Logistic Regression
C	1
class_weight	None
dual	FALSE
fit_intercept	TRUE
intercept_scaling	1
l1_ratio	None
max_iter	100
multi_class	auto
n_jobs	None
penalty	l2
random_state	42
solver	lbfgs
tol	0.0001
verbose	0
warm_start	FALSE

Appendix E
Optimized Logistic Regression Hyperparameters

	Optimized Logistic Regression
C	0.3148911648
class_weight	None
dual	FALSE
fit_intercept	TRUE
intercept_scaling	0.7322619479
l1_ratio	0.5990598257
max_iter	1000
multi_class	auto
n_jobs	None
penalty	l2
random_state	42
solver	lbfgs
tol	0.0001
verbose	0
warm_start	FALSE

Appendix F
Light Gradient Boosting Machine Hyperparameters

	Light Gradient Boosting Machine
boosting_type	gbdt
class_weight	None
colsample_bytree	1
importance_type	split
learning_rate	0.4
max_depth	-1
min_child_samples	66
min_child_weight	0.001
min_split_gain	0
n_estimators	230
n_jobs	-1
num_leaves	80
objective	None
random_state	42
reg_alpha	0.0005
reg_lambda	0.05
subsample	1
subsample_for_bin	200000
subsample_freq	0
feature_fraction	1
bagging_freq	6
bagging_fraction	0.9

Appendix G
Ridge Classifier Hyperparameters

	Ridge Classifier
alpha	6.78
class_weight	None
copy_X	TRUE
fit_intercept	TRUE
max_iter	None
positive	FALSE
random_state	42
solver	auto
tol	0.0001

APPENDIX H

Extreme Gradient Boosting Classifier Hyperparameters

	Extreme Gradient Boosting Classifier
objective	multi:softprob
base_score	None
booster	gbtree
callbacks	None
colsample_bylevel	None
colsample_bynode	None
colsample_bytree	0.9
device	cpu
early_stopping_rounds	None
enable_categorical	FALSE
eval_metric	None
feature_types	None
gamma	None
grow_policy	None
importance_type	None
interaction_constraints	None
learning_rate	0.3
max_bin	None
max_cat_threshold	None
max_cat_to_onehot	None
max_delta_step	None
max_depth	5
max_leaves	None
min_child_weight	3
missing	None
monotone_constraints	None
multi_strategy	None
n_estimators	290
n_jobs	-1
num_parallel_tree	None
random_state	42
reg_alpha	1
reg_lambda	0.0005
sampling_method	None
scale_pos_weight	31.4
subsample	0.3
tree_method	auto
validate_parameters	None
verbosity	0

APPENDIX I

Extreme Gradient Boosting Classifier Hyperparameters

	Stacked Classifier
Model 1	Light Gradient Boosting Machine
Model 2	Ridge Classifier
Model 3	Extreme Gradient Boosting Classifier